

Adding & Subtracting Polynomials — Practice Worksheet

Name: _____ Date: _____ Period: _____

Type 1 — Adding Polynomials (Find the Sum)

EXAMPLE — Look at this before you begin:

Find the sum: $(5ab + 5ac - 9bc) + (-ab + 10ac + 4bc)$

Group like terms: $(5ab - ab) + (5ac + 10ac) + (-9bc + 4bc)$

Simplify: $4ab + 15ac - 5bc$

Answer: $4ab + 15ac - 5bc$. Show all work!

Find each sum. Show all work. Circle your final answer.

Problem 1-A. Find the sum.

$$(3xy + 2xz - 4yz) + (-xy + 5xz + 2yz)$$

Problem 1-B. Find the sum.

$$(7ab - 3ac + bc) + (2ab + 6ac - 5bc)$$

Problem 1-C. Find the sum.

$$(4m^2n - 3mn^2 + 2mn) + (-m^2n + 5mn^2 - 7mn)$$

Problem 1-D. Find the sum.

$$(8x^2y + 3xy^2 - xy) + (-2x^2y - xy^2 + 4xy)$$

Problem 1-E. Find the sum.

$$(-5a^2b + 4ab^2 - 3ab) + (3a^2b - 2ab^2 + 8ab)$$

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Type 2 — Subtracting Polynomials (Find the Difference)

EXAMPLE — Look at this before you begin:

Find the difference: $(-12r^2s + 2rs^2 + 6rs) - (3r^2s + 8rs^2 - rs)$

Distribute the negative: $-12r^2s + 2rs^2 + 6rs - 3r^2s - 8rs^2 + rs$

Combine like terms: $-15r^2s - 6rs^2 + 7rs$

Answer: $-15r^2s - 6rs^2 + 7rs$. Show all work!

Find each difference. Show all work. Circle your final answer.

Problem 2-A. Find the difference.

$$(6x^2y - 3xy^2 + 5xy) - (2x^2y + xy^2 - 3xy)$$

Problem 2-B. Find the difference.

$$(8ab^2 - 5a^2b + 3ab) - (-2ab^2 + 4a^2b + ab)$$

Problem 2-C. Find the difference.

$$(5m^2n - 3mn + 2mn^2) - (2m^2n + mn - 4mn^2)$$

Problem 2-D. Find the difference.

$$(-4r^2s + 7rs^2 - 2rs) - (3r^2s - 2rs^2 + 5rs)$$

Problem 2-E. Find the difference.

$$(9x^2z - 4xz^2 + 6xz) - (-x^2z + 2xz^2 - 3xz)$$

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Answer Key

Type 1 — Adding Polynomials

- 1-A $2xy + 7xz - 2yz$
- 1-B $9ab + 3ac - 4bc$
- 1-C $3m^2n + 2mn^2 - 5mn$
- 1-D $6x^2y + 2xy^2 + 3xy$
- 1-E $-2a^2b + 2ab^2 + 5ab$

Type 2 — Subtracting Polynomials

- 2-A $4x^2y - 4xy^2 + 8xy$
- 2-B $10ab^2 - 9a^2b + 2ab$
- 2-C $3m^2n - 4mn + 6mn^2$
- 2-D $-7r^2s + 9rs^2 - 7rs$
- 2-E $10x^2z - 6xz^2 + 9xz$